FRANCESCO GANIS

PhD, Sound & Music Engineer

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With a strong background in audio digital signal processing (DSP), prototyping, and the evaluation of audio solutions, I bring experience in developing user-centered audio technologies and vibrotactile feedback systems. My work has involved translating user and system needs into structured designs, defining experimental approaches, and iterating on prototypes based on data and user feedback. I am a passionate audio engineer and researcher, with deep love for music and sound. I am eager to contribute, while continuing to expand my knowledge and skills. You can find my portfolio at francescoganis.github.io.

EDUCATION

PhD, Tangible Interactive Technologies for Hearing and Music Development

Aalborg University

Sep. 2022 – Mar. 2026 | Copenhagen, DK

Skills learned: Research | Prototyping | Evaluating

- Designed and evaluated interactive audio and vibrotactile systems for children with hearing impairment
- Developed experimental approaches to assess system performance, user interaction, and training outcomes
- Translated user and clinical needs into system requirements, prototypes, and iterative design improvements
- Collaborated with audiologists and clinicians to align technical development with real-world use contexts
- Produced scientific and technical documentation of methods, system behavior, and results

Master's Degree, Sound & Music Computing

Aalborg University

Sep. 2020 – Jun. 2022 | Copenhagen, DK

Skills learned: DSP | Prototyping | Evaluating

- Thesis: "Vibrotactile Feedback for Musical Training of Cochlear Implants Users"
- Completed courses in Sound Processing, Embodied Interaction, Music Perception and Cognition, New Interfaces for Musical Expression

Bachelor's Degree, Electronic Music & Sound Engineering

"C. Pollini" Conservatory

Sep. 2016 – Apr. 2020 | Padova, IT

Skills learned: Sound Engineering | MAX/MSP | Sound Design

- Graduated with honors
- Thesis: "Continuous Sonification of Autonomous Driving Vehicles with Purposes of Passengers Information and Trust Augmentation"

STRENGTHS

Technical Skills

MATLAB | Prototyping | DSP | Product Evaluation | Audio Testing | C | Arduino | Git | Unity | C# | Embedded Programming | ESP32 | LaTeX | Lasercutting | MAX/MSP | Pro Tools | Pure Data | Reaper | Sound Engineering | 3D Printing

Soft Skills

Analytical Thinking | Collaboration | Detail-oriented | Problem-solving | Scientific Dissemination | Supervision | Teamwork | Technical Documentation

OTHER EXPERTISE

Photography | Playing Drums

LANGUAGES

English ●●●●●
Full professional proficiency

Italian ●●●●●
Mother tongue

PUBLICATIONS

Thesis

- F. Ganis, "Tune in: Exploring music and sound training for children with hearing challenges," Ph.d.-serien for Det Tekniske Fakultet for IT og Design, Aalborg Universitet, 2025.

EXPERIENCE

Internship, Technology Evaluation Engineer

Bang & Olufsen

Apr. 2026 – May 2026 | Lyngby, DK

Skills learned: Product Evaluation | Perceptual Testing | Technical Documentation

- Development of experimental paradigm for the perceptual evaluation of high-end audio products
- Defined evaluation criteria, relevant attributes, and study procedures to support structured, evidence-based assessment
- Documented and communicated findings through technical reporting and presentations for internal stakeholders

Teaching Assistant, Sound & Music Computing

Aalborg University

Sep. 2022 – Aug. 2025 | Copenhagen, DK

Skills learned: Teaching | Communication | Supervision

- Supported teaching in the *Sound Processing* and *Physical Models for Sound Synthesis* courses
- Supervised student projects in the Medialogy and Sound & Music Computing programmes

Student Helper, Sound Synthesis and Design

Aalborg University

Oct. 2021 – Apr. 2022 | Copenhagen, DK

Skills learned: MATLAB | Audio Programming | DSP

- Created sound examples in MATLAB for the book *Serafin, S., Buxton, B., Gaver, B., & Bly, S. (2022). Auditory interfaces. Focal Press*

Internship, Prototype Development

Oticon Medical

Oct. 2021 – Jan. 2022 | Copenhagen, DK

Skills learned: Product Development | Prototyping | Evaluation

- Developed a vibrotactile smartphone prototype to support music listening for users with hearing impairment
- Translated user needs into functional prototype features in a hearing-technology setting
- Evaluated the system with attention to usability and perceptual experience

Sound Engineer, Studio and Live Sound Engineering

Various

2017 – 2018 | Veneto Region, IT

Key areas: Sound Engineering | Live Electronics | Audio Production

- Worked across studio and live sound contexts, including live electronics, editing, and front-of-house support

Journal Articles

- F. Ganis et al., "The role of haptics in training and games for hearing-impaired individuals: A systematic review," *Multimodal Technologies and Interaction*, vol. 8, no. 1, p. 1, 2023.

Conference Proceedings

- F. Ganis et al., "Vibrotactile memory: A case study of timbre perception training in children with cochlear implants using a video game," SMC Proceedings, 2024, pp. 134–141.
- F. Ganis et al., "Tickle tuner-haptic smartphone cover for cochlear implant users' musical training," in *HAID*, Springer Nature, 2022, p. 14.
- F. Ganis et al., "Real-time timbre transfer and sound synthesis using ddsp," SMC Proceedings, 2021, pp. 175–182.

FURTHER TRAINING

Masterclass, Live Electronics

Chigiana Academy

Jul. 2019 | Siena, IT

Masterclass, Symphonic Orchestra Recording

G. A. Fano Institute

Jul. 2018 | Spilimbergo, IT

Diploma, Music, Theory and Solfège

G. Tartini Conservatory

Sep. 2009 | Trieste, IT

REFERENCES

Prof. Stefania Serafin
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@ Righospitalet, Copenhagen - DK
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Prof. Nicola Bernardini
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